Esteban Dominguez

Faithful Hard Worker

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Portfolio

EXPERIENCE

Tufts University Mechanical Engineering Department, Medford, MA 02155 — *Materials & Manufacturing Lab Teaching Assistant*

Sep 2026 - Present

- Help students manufacture parts across a range of processes, including a sheet metal box, a sand-cast pin, and a part machined directly from an engineering drawing
- Operate a drill press, mill, and lathe daily while guiding students through proper technique
- Run heat transfer and CFD simulations in COMSOL Multiphysics to support design analysis

Vida Life Church, Mesquite, TX — *Volunteer: AV Technician, Kids Church Leader & Jr. Commander*

Aug 2022 - Aug 2024

- Ran live audio and video for the weekly English service, including sound mixing, camera operation, and livestream setup
- Led activities for Kids Church, ages 5–12, combining games with weekly lessons
- Served as Jr. Commander in Royal Rangers, planning and teaching weekly lessons to boys ages 6–8

Tufts Dining, 200 Packard Ave, Medford, MA 02155 — *Dining Assistant*

Aug 2024 - May 2025

- Maintained cleanliness and setup of the dining room, including sanitizing and arranging tables and chairs
- Sorted and restocked dishware in the dishroom to keep service running smoothly
- Operated the pizza station, preparing food ahead of the lunch rush

Bright Management, Dallas, TX — *Professional Fundraiser*

June 2025 - June 2025

- Approached and engaged the public to represent and raise support for a charity
- Developed concise, persuasive communication skills under time pressure

TECHNICAL SKILLS

3D Modeling & CAD (Inventor, SolidWorks, some Onshape)

COMSOL Multiphysics (Heat Transfer & CFD)

SolidWorks Simulation (Static FEA) & Motion Studies

MATLAB & Python for engineering analysis

3D Printing & Rapid Prototyping

Proficient in VS Code

Engineering Drawings & GD&T

Basic Networking & Server Maintenance (built and maintain a home NAS)

Proficient with Computers & AI Tools (local and cloud-based)

SOFT SKILLS

Hard worker

Focused

Great Communicator

Works well in groups

Patient with people

Quick learner

AWARDS

Dean's List (Fall 2024, Spring 2025, Fall 2025, Spring 2026)

AP Scholar Award

SPOKEN LANGUAGES

English (native), Spanish (fluent), French (basic)

PROGRAMMING

MATLAB, Python, Java, JavaScript, SystemVerilog

EDUCATION

SEM at Townview, Dallas, TX — High School Diploma

Aug 2020 - May 2024

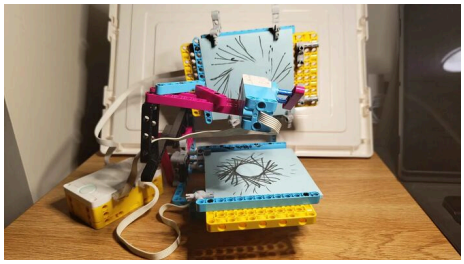
Graduated with AP Honors from a selective magnet program.

Tufts University, Medford, MA — B.S. in Mechanical Engineering

Sep 2024 - Expected May 2028 • GPA: 3.44

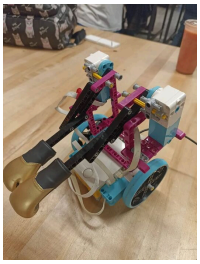
Relevant coursework: Mechanics I & II, Thermal Fluid Systems I & II, Materials & Manufacturing I & II, Engineering Design I & II, Electronics & Controls I, Intro to Digital Electronics, Aerodynamics.

PROJECTS



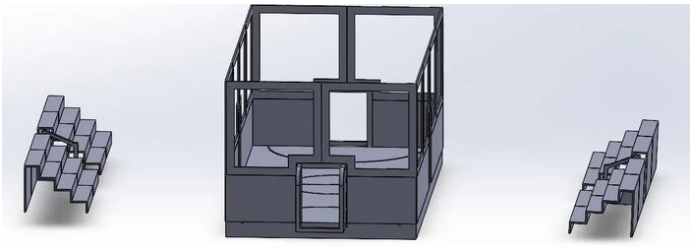
LEGO Spike Prime Drawing Machine

A two-axis pen plotter built from LEGO Spike Prime, using motor-driven linkages to trace geometric patterns such as spirals and radial bursts.



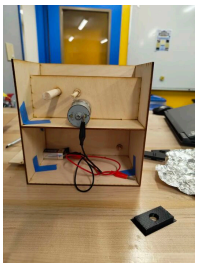
LEGO Spike Prime Boxing Robot

A self-balancing, two-wheeled robot with boxing-glove end effectors, controlled in real time by laptop camera hand-tracking so a player's punches drive the robot's. Built with a team, combining mechanical design, motor control, and computer vision.



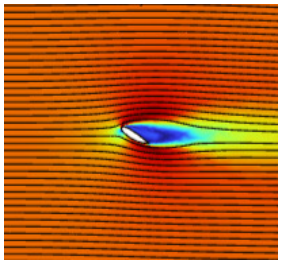
Futsal Ring

A full-scale futsal rink modeled in CAD, then manufactured using sheet metal bent on a brake and laser-cut acrylic for the center circle, plus a matching set of game pucks.



Motorized Crank Mechanism

A 4-bar linkage embosser designed in CAD, then built as a working prototype with a laser-cut wood housing and a DC motor driving the linkage.



2D Airfoil CFD Simulation (COMSOL)

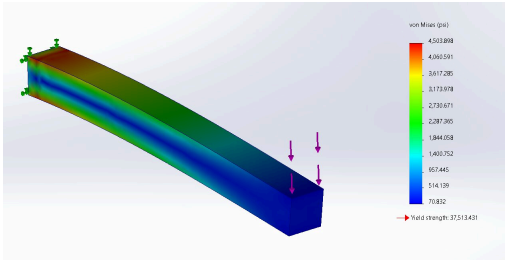
Laminar flow around a custom airfoil at a 26.6° angle of attack (Re ≈ 437). Ran a mesh-refinement study on coarse, normal, and fine meshes, pulled lift and drag from the pressure and viscous forces, and compared the results with an instructor benchmark in a technical report.

```
// first-order power estimate
P = ρ · Q · g · H
  = 1000 · 10 · 9.81 · 4.4 ≈ 432 kW
P_real = 0.75 · P ≈ 324 kW

Archimedes screw · Ø 5 m
H = 4.4 m · Q = 10 m³/s
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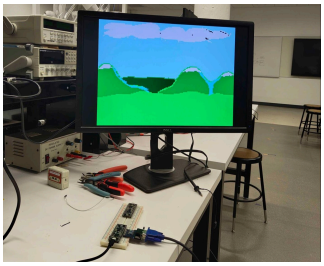
Charles River Hydroelectric Study

Picked the Silk Mill Dam site on the Charles River for its steady flow and 4.4 m head. Sized a 5 m Archimedes screw turbine and built a first-principles power model, estimating about 324 kW of real output.



Cantilever Beam FEA (SolidWorks Simulation)

Static FEA of an end-loaded cantilever beam, validated against beam theory: 0.019 in max deflection by hand and in simulation, and 4.5 vs. 4.503 ksi max bending stress. Checked peak stress against the 37.5 ksi yield strength.



FPGA Modulus Math Game (Team of 4)

A timed arithmetic game on two FPGAs in SystemVerilog: one reads a 4×4 keypad, the other runs the game logic and drives a 640×480 VGA display. Created the pixel-art sprites and backgrounds and helped verify the LFSR random number generator.